

8 (**short**) A spinning firework is made from a uniform disk of radius a and mass M that is free to rotate about a pin through its central axis, as shown in Fig. 5. Two rockets are mounted at the edge of the disk such that they fire in opposite directions, causing the disk to spin.

(a) By modelling each rocket as a thin rod of length $3a/2$ and mass $M/2$, calculate the mass moment of inertia of the firework. [3]

(b) Each rocket ejects mass at a constant rate $\dot{m}$ with relative velocity v . Ignoring any effect of gravity, calculate the initial angular acceleration of the firework. [4]

(c) Assuming that air resistance is negligible, that the ejected mass is removed uniformly along the rods, and that their whole mass can be ejected, sketch the variation of the angular acceleration with time. [3]

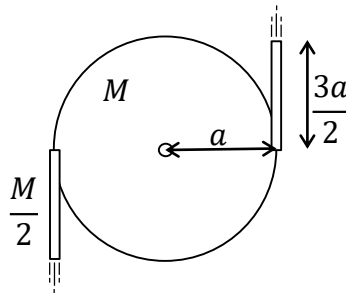


Fig. 5